Cohort canonical evaluation — 10 agents

idea-01-lateral-attribution · 130 held-out segments · V0 yaw 0.01456 rad/s, CTE 147.44 m · generated 2026-05-31 21:53

Executive summary

AGENTS (OK / TOTAL)

10 / 10

EVAL POOL

130 segs

V0 YAW RMSE

0.01456

V0 CTE RMSE

147.44 m

WALL TIME

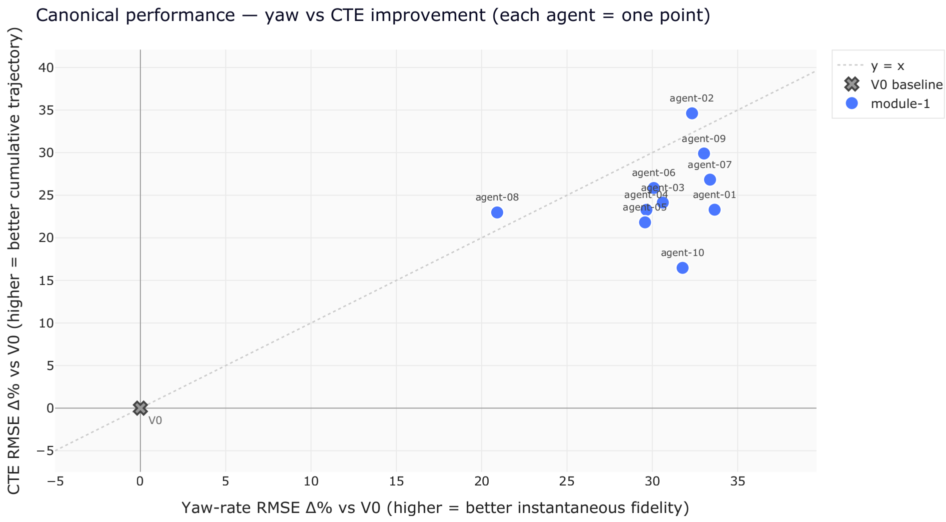
3.82s

Best yaw: `m1-agent-01` **+33.6%** · Best CTE: `m1-agent-02` **+34.6%**

Winning both KPIs ≥ +30% (1 agents): `m1-agent-02`

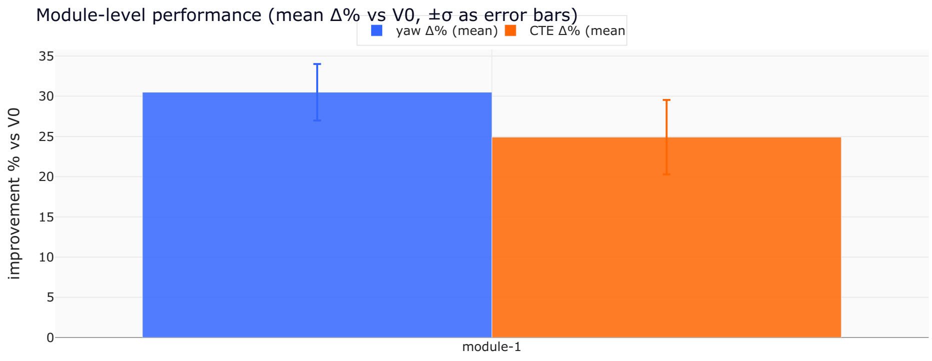
1. Headline scatter — yaw vs CTE improvement

Each agent is one point. **Top-right is the goal** — both KPIs improve over V0. The dashed diagonal is y = x: above it, an agent improved CTE more than yaw (good trajectory integration); below, the opposite. The V0 baseline sits at the origin.



2. Module-level aggregation

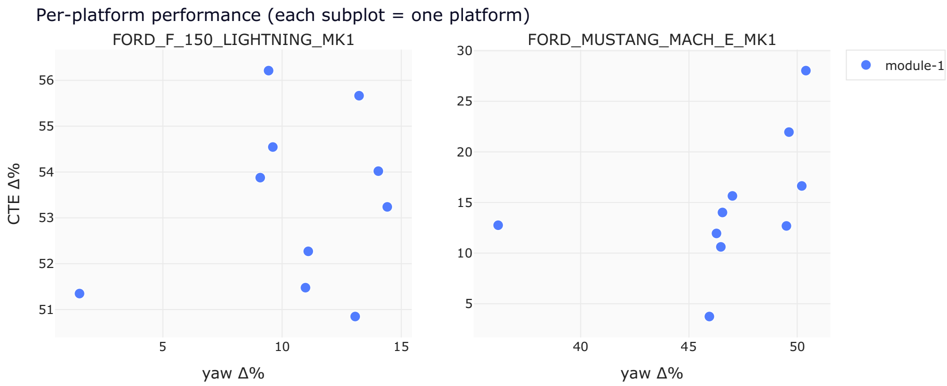
Performance grouped by which module shipped the agent. Mean Δ% over only the agents that reconstructed successfully; failures shown separately.



family	n ok / total	yaw Δ% (mean ± σ)	CTE Δ% (m
module-1	10/10	+30.5% ± 3.5%	+24.9% ±

3. Per-platform breakdown

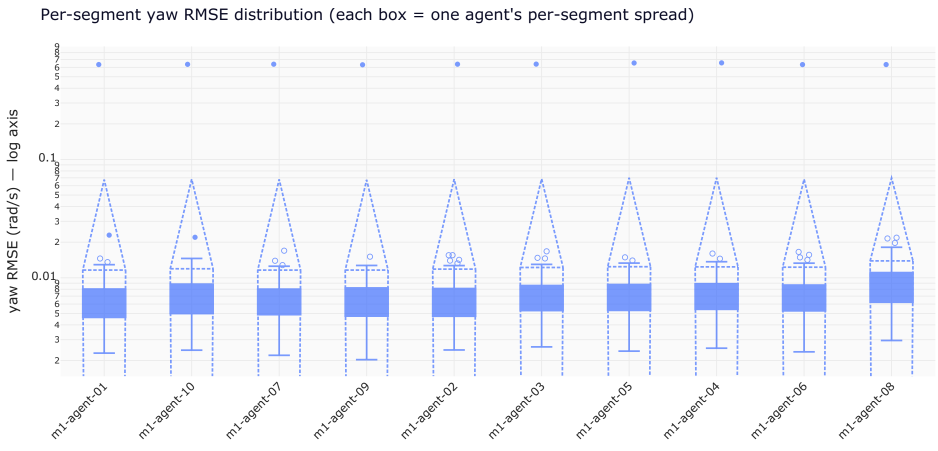
Same cohort, but split by the vehicle platform the agent was evaluated on. Agents that win one platform but lose another have a platform-specific calibration; agents whose dots cluster together generalise.



platform	agents	yaw Δ% (mean ± σ)
FORD_F_150_LIGHTNING_MK1	10	+10.6% ± 3.6%
FORD_MUSTANG_MACH_E_MK1	10	+46.8% ± 3.9%

4. Per-segment yaw RMSE distribution

Each box summarises one agent's per-segment yaw RMSE — pooled RMSE can hide that an agent does great on most segments but pathologically badly on a few. The y-axis is logarithmic to compress outliers.



5. Per-agent canonical scorecard

The full table. Failed reconstructions are surfaced with their reason — no fake zeros.

agent	family	status	yaw V0	yaw final
m1-agent-01	module-1	ok	0.014563	0.009662
m1-agent-02	module-1	ok	0.014563	0.009855
m1-agent-03	module-1	ok	0.014563	0.010105
	module-1	ok	0.014563	0.010244

agent	family	status	yaw V0	yaw final
m1-agent-04				
m1-agent-05	module-1	ok	0.014563	0.010256
m1-agent-06	module-1	ok	0.014563	0.010181
m1-agent-07	module-1	ok	0.014563	0.009701
m1-agent-08	module-1	ok	0.014563	0.011518
m1-agent-09	module-1	ok	0.014563	0.009753
m1-agent-10	module-1	ok	0.014563	0.009936

6. Calibration cards — fitted coefficients

Where the cohort converges on similar physics, and where it forks. Each tile is one agent's `coeffs.json` as shipped.

m1-agent-02


```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.4760851016953556,
    "K_us": 0.0026722138032651867,
    "bias_rad": 0.0002686192499419401
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.809760457416018,
    "K_us": 0.0038412171248315657,
    "bias_rad": 0.001307379955568698
  },
  "HYUNDAI_IONIQ_5": {
    "L": 3.096229822470566,
    "K_us": 0.003695564361835472,
    "bias_rad": -0.0002960976826300522
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "K_us": 0.0,
    "bias_rad": 0.0
  }
}
```


m1-agent-04


```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "a": -0.004329037631466362,
    "b": 0.9177897626357783,
    "c": -0.00044178131403551616
  },
  "FORD_MUSTANG_MACH_E_MK1": {
```

```
    "a": 0.0002206757945255914,
    "b": 1.1513559927303039,
    "c": -0.0005583363495612761
  },
  "HYUNDAI_IONIQ_5": {
    "a": 0.002022989265629857,
    "b": 0.9120045284217976,
    "c": -0.00048449411444108955
  },
  "TESLA_MODEL_3": {
    "a": 0.0002206757945255914,
    "b": 1.1513559927303039,
    "c": -0.0005583363495612761
  },
  "_default": {
    "a": 0.0,
    "b": 1.0,
    "c": 0.0
  }
}
```

m1-agent-06

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
```

```
    "best": "M2_understeer",
    "m1": {
      "a": 1.0870612234849137
    },
    "m2": {
      "L_eff": 2.4763030468069056,
      "K_us": 0.0024395167795073017
    },
    "m3": {
      "a": 0.9751496691433119,
      "b": 0.2383949727329036,
      "c": 0.0004082870644824788
    },
    "val_rmse": {
      "V0": 0.0267999381028469,
      "M1": 0.027439581036525294,
      "M2": 0.026420879339395156,
      "M3": 0.031903168689021696
    },
    "train_rmse": {
      "V0": 0.012685628695945105,
      "M1": 0.011389013310887795,
      "M2": 0.008745921115146504,
      "M3": 0.010386863485176999
    }
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "best": "M2_understeer",
    "m1": {
      "a": 0.8569324797501493
    },
    "m2": {
      "L_eff": 3.882198880475702,
      "K_us": 0.003439279945738158
    },
    "m3": {
      "a": 0.8772240685582882,
      "b": -0.031353830652592465,
      "c": -0.0036354381262862774
    },
    "val_rmse": {
      "V0": 0.01591104278075153,
```

```
      "M1": 0.011141896721881754,
      "M2": 0.0073696198273352906,
      "M3": 0.0104650790814036
    },
    "train_rmse": {
      "V0": 0.02018452928972004,
```

m1-agent-08

```
{
  "HYUNDAI_IONIQ_5": {
    "L": 3.0,
    "K": 0.0033895612010179177,
    "s": 0.9668278221152291,
    "off": 0.0005045046778856075,
    "lag": 3,
    "rmse": 0.00888482422851739
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "K": 0.002625595512351926,
    "s": 1.1771921499482927,
    "off": 3.9923690213217876e-05,
    "lag": 4,
    "rmse": 0.01315660341959157
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "K": 0.0035388455508032707,
    "s": 0.9569361755429138,
    "off": -0.0011908292677060655,
    "lag": 3,
    "rmse": 0.011832340064259452
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "K": 0.0033895612010179177,
```

```
    "s": 1.0,
    "off": 0.0,
    "lag": 3,
    "rmse": null
  }
}
```


m1-agent-10


```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L_eff": 2.5297190644305063,
    "K_us": 0.0015446536838391765,
    "d0": -4.756112262525493e-05,
    "tau_yaw": 0.02,
    "tau_steer": 0.05
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L_eff": 3.834827559665228,
    "K_us": 0.0031524889056087045,
    "d0": 0.00115199881391885,
    "tau_yaw": 0.0,
    "tau_steer": 0.05
  },
  "HYUNDAI_IONIQ_5": {
    "L_eff": 3.1485752690608293,
    "K_us": 0.0021732037046845788,
    "d0": -0.0005619614330402095,
    "tau_yaw": 0.0,
    "tau_steer": 0.05
  },
  "TESLA_MODEL_3": {
```

```
    "_note": "no truth available \u2014 passthrough yaw_rate_pred_
    "passthrough": true
  }
}
```

6b. Operating-contract audit

The grader strips `sim_df` to an allowlist before calling each agent's `predict()`, so an agent cannot read truth at inference time. This section reports source-level intent — which agents *would have* read truth columns if the grader hadn't intervened. A clean audit is its own quality signal.

> **Clean cohort:** no agent referenced truth-derived column names inside their predict() body.

7. Reconstruction quality (substrate signal)

How many agents shipped the right artefacts to be canonically gradable. Failures here are a substrate or contract problem, not a model problem.

format check	pass	fail
agent_folder_exists	10	0
has_manifest_json	10	0
manifest_parsable	10	0
manifest_declares_predict_callable	10	0
manifest_declares_platform_support	10	0
has_predict_py	10	0
has_coeffs_json	9	1
has_report	0	10

8. Worst-of-cohort

Lowest yaw Δ%

- m1-agent-08 (+20.9%)
- m1-agent-05 (+29.6%)
- m1-agent-04 (+29.7%)

Lowest CTE Δ%

- m1-agent-10 (+16.5%)
- m1-agent-05 (+21.8%)
- m1-agent-08 (+23.0%)